

Course name : Human-computer interaction

Academic year : 2025-2026

Ինֆորմատիկայի բակալավր (հայկական)/Licence en Informatique de l'Université de Toulouse (ֆրանսիական)		
Qualification : Computer Science		
Coordinating faculty / chair : IMA		
Year of study : 3 Semester : 1		3.0 ECTS
Total hours: 21.0	Lecture (CM): Tutorials (TD) : 9.0 Lecture and Tutorials CTD : Practicals (TP) : 12.0 Lecture and Practicals (CTP): Project:	
Estimated number of hours of individual student work (TPS): 60.0		
Language of instruction : English		
Instructor (TD) : Gyulgyulyan Eliza Instructor (TP) : Gyulgyulyan Eliza		
Instructor's email : Eliza.Gyulgyulyan@pers.ufar.am		

PURPOSE AND DESCRIPTION OF THE COURSE

This course introduces the theory, methods and practice of Human–Computer Interaction (HCI) for third-year Computer Science bachelor degree students. Students will explore cognitive, perceptual and social factors that influence interaction design, practice user-centred design methods, develop prototypes using current industry tools, and carry out usability evaluation. The course balances conceptual foundations (models, metrics, and ethics) with hands-on design and coding work so students graduate with both critical thinking and practical prototyping skills.

INTENDED LEARNING OUTCOMES

Knowledge and skills

A - Knowledge	A1.1. Describe databases, human-computer interaction, elements of artificial intelligence, graphical information, and parallel programming issues., (N3) Niveau de Maîtrise A1.2. Know the basic principles of Human-Computer Interaction & User-Centered Design, (N2) Niveau d'Application A1.3. Master the principles of the of design and modeling of interactive systems, (N2) Niveau d'Application
B1 – Skills to apply professional knowledge	B1.1. Analyze and formulate problems, then design and implement solutions by selecting appropriate software development tools., (N2) Niveau d'Application B1.2. Implement interactive software prototypes using industrystandard tools (e.g., Balsamiq, Axure) that adhere to fundamental HCI principles and user-centered design methodologies., (N2) Niveau d'Application B1.3. Evaluate the usability and effectiveness of existing interactive systems, identifying areas for improvement and proposing concrete, HCI-driven design modifications., (N2) Niveau d'Application B1.4. Develop a comprehensive software architecture for the interactive components of a given application, justifying design decisions based on established HCI theories and best practices., (N2) Niveau d'Application
B2 – General (transversal) skills	B2.1. Utilize professional literature in English., (N2) Niveau d'Application B2.2. Apply written and oral professional communication in English., (N2) Niveau d'Application B2.3. Collaborate effectively in teams to analyze user needs, conceptualize design solutions, and present persuasive arguments for user-centered interactive systems., (N2) Niveau d'Application B2.4. Communicate complex technical and design concepts related to human-computer interaction to both technical and non-technical audiences, utilizing appropriate visual and written documentation., (N2) Niveau d'Application

KNOWLEDGE / SKILLS ASSESSMENT & EVALUATION			
Form of assessment	Ongoing evaluation ☒	Midterm exam ☐	Final exam ☐
Exam type	Oral ☐ Written ☒	Oral ☐ Written ☐	Oral ☒ Written ☒

Exam task	3 short practicals and 2 design critique reports. Graded on HCI reasoning, application of principles, clarity and reflection.		Group project (3-5 students): working prototype, usability study report, project report and 5min presentation. Graded on design rationale, usability evaluation, technical implementation & teamwork.
Group work	yes ☐ no ☒	yes ☐ no ☐	yes ☒ no ☐
Duration: (hours)	0.5		2
Weight in final grade	30%		70%

(100%)			
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Assessment Criteria:	The ongoing assessment will be a 20-point individual assessment. 5 points will be allocated to two written design reports in which students will analyze existing products (hardware and software), applying specific HCI principles to critique their user interface and justify their findings. The remaining 15 points will be for 3 short practical, inclass exams. During the exam, students will be given a small design problem and will have to produce a lowfidelity prototype by filling in an existing code portion and explaining their solution		course group project with 35 students and a total weight of 20 points, the assessment is divided into a report (5 points), video presentation (5 points), and a defense (10 points). The report's weight is distributed as follows: 2.5 points for problem and user analysis, 2.5 points for the design process and prototype. The defense, with a duration of 5 minutes per group, is assessed on 5 points for the quality of the presentation (including clarity, time management, and group cohesion) and 5 points for the group's depth of understanding, demonstrated through their answers during the Q&A session and their ability to justify design decisions based on core HCI principles. The entire project grading will consider the following criteria: <i>Does the design intervention address a real population and/or situation in Armenia (at least)?</i> • <i>Does the intervention use technology in an appropriate and novel way?</i> • <i>Was relevant prior work properly identified and cited?</i> • <i>Did the team explore the entire ecosystem of stakeholders,</i>
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			<i>conditions, and contexts?</i> ☐ <i>Was the intervention wellcrafted and effectively presented?</i>
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TEACHING METHODS AND TOOLS

Lectures: brief, concept-driven introductions.

Tutorials: guided exercises and discussion.

Practicals/Labs: hands-on prototyping, coding and usability testing.

Design Critiques: peer feedback cycles.

Project-based learning: team projects with milestone submissions.

Tools: Balsamiq/Figma/Axure/HTML/CSS/JavaScript for interactive prototypes, Java for lab exercises,

Git/GitHub or Google Forms for usability surveys.

KNOWLEDGE & SKILLS PREREQUISITS

Students should have: basic programming skills (Java or other high-level language), basic statistics knowledge, familiarity with software development lifecycle and fundamental data structures. Prior coursework in programming and software engineering is recommended.

COURSE SYLLABUS AND RESOURCES

Topic	Hours	Core resources	Additional resources
TD			
1. Introduction to HCI The Human	1,5	Norman, D. A., 2013, The design of everyday things Revised and expanded edition. Pages 1-122	Alan Dix, Janet Finlay, Gregory D. Abowd, Russell Beale, HumanComputer Interaction pages 1-6

2. The Computer, Fitts' Law	3	Norman, D. A., 2013, The design of everyday things Revised and expanded edition. Pages 123-216	https://www.interaction-design.org/literature/article/fitts-law-tracking-users-clicks https://www.interaction-design.org/literature/article/fitts-law-tracking-users-clicks Alan Dix, Janet Finlay, Gregory D. Abowd, Russell Beale, Human- Computer Interaction
			pages 7-14
3. The interaction Frameworks and HCI Ergonomics	1.5	Alan Dix, Janet Finlay, Gregory D. Abowd, Russell Beale, HumanComputer Interaction, 2005 pages 15-21	
4. Cognitive Modeling Dialog Notations and design Models of System	3	Alan Dix - Humancomputer interaction Pearson Education (2004) pp.419,544,594	
TP			
1. Design Critiques	1.5	A good example of the assignment	
2. Automata and STNs	1.5	A good example of the assignment	
3. Java Coding with switch cases	3	A good example of the assignment	

4. DESIGN THINKING Problem analysis and definition Prototyping Evaluation Designing for interaction demonstration	4.5	Design Thinking, Gavin Ambrose, Paul Harris, AVA Publishing SA 2010, Degen, H., & Ntoa, S. (Eds.). (2023). Artificial Intelligence in HCI. Springer.	Design Thinking Handbook by Eli Woolery, design better,co by Invision, Greenberg, S., Carpendale, S., Marquardt, N., Buxton, B. (2021). Sketching User Experiences: The Workbook (2nd ed.). Morgan Kaufmann.
5. Presentation	1.5		

CORE BIBLIOGRAPHY

Core textbooks, publications, and research articles:

1. Buxton, W., A three-state model of graphical input. IFIP TC13 Conference on HCI, North-Holland Publishing Co., 1990
2. Design Thinking, Gavin Ambrose, Paul Harris, AVA Publishing SA 2010
3. Design Thinking Handbook by Eli Woolery, designbetter,co by Invision
4. Alan Dix - Human-computer interaction -Pearson Education (2004)

Supplementary books, textbooks, publications, and research articles :

1. Bødker, S. (2022). HCI and Design Thinking: From Interaction to Practice. MIT Press.
2. Hartson, R., & Pyla, P. S. (2024). *The UX Book: Agile UX Design for a Quality User Experience* (3rd ed.). Morgan Kaufmann
3. Hooper, Steven_Berkman, Eric - Designing mobile interfaces-O'Reilly (2012)
4. Rogers, Y., Sharp, H. (2023). *Interaction Design: Beyond Human-Computer Interaction* (6th ed.). Wiley
5. Greenberg, S., Carpendale, S., Marquardt, N., Buxton, B. (2021). *Sketching User Experiences: The Workbook* (2nd ed.). Morgan Kaufmann.
6. Hooper, Steven_Berkman, Eric - Designing mobile interfaces-O'Reilly (2012)
7. [Literature source Silva, T., & Müller, F. \(2024\). AI in Human-Computer Interaction: Emerging Practices. ACM Press.](#)

Web resources :

1. <https://www.nngroup.com/articles/design-thinking/>
2. <https://www.interaction-design.org/literature/article/fitts-law-tracking-users-clicks>

3. https://drive.google.com/file/d/1rDjArw_AI_15NKVfAhh-OeK0POMdNCCP/view?usp=sharing
4. <https://uizard.io>
5. <https://www.midjourney.com>